

Sreevaatsav Bavana

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Personal Profile

Gold medalist in B.Tech Artificial Intelligence at Mahindra University, currently working as Senior Data Scientist at Penguin AI. Strong expertise in Multimodal AI, LLM systems, and Machine Learning, with research interests in learning robotic manipulation policies from multimodal perception, sim-to-real transfer, and grounded, interpretable vision-language models. Passionate about building AI systems that bridge perception, language, and real-world action.

Education

Mahindra University, Hyderabad, India

Aug 2020 - Aug 2024

Bachelor of Technology in Artificial Intelligence

- **Gold Medalist** of the 2020–2024 B.Tech AI batch
- **Cumulative GPA** : 8.76
- **Relevant Courses**: Data Structures and Algorithms, Linear Algebra, Probability and Statistics, Optimization Techniques in AI, DBMS, Machine and Deep Learning, Natural Language Processing, Reinforcement learning

Publications

GIV-CXR: Densely Grounded, Visually Interpretable, Chest X-Ray Question Answering Dataset

2026

Sreevaatsav Bavana, Adit Rushil Potta, Sai Amrit Patnaik, Nidhi Goyal

MedReasoner Workshop at CVPR

From Layoff to Reemployment: Analyzing Social Media Narratives

2025

Yayati Gupta, Sreevaatsav Bavana, Sudhira Chegu, Sujith Sai Kalakonda, Yash Mathur, Manmitha Polsani, Sanatan Sukhija

Computational Data and Social Networks (CSoNet 2025)

A Systematic Study on Video Summarization: Approaches, Challenges, and Future Directions

2023

Kajal Kansal, Nikita Kansal, Sreevaatsav Bavana, Bodla Krishna Vamshi, Nidhi Goyal

Proceedings of the 2nd Workshop on User-centric Narrative Summarization of Long Videos

Projects

Imitation-Learning Policies on a Fairino FR5 Cobot

Jun 2026 - Present

Independent robotics research

- Collected 1,021 teleoperated demonstrations (~11.5 hours) on a 6-DOF Fairino FR5 across three manipulation tasks — colour sorting, sanding, and pick-to-light — recorded with a wrist-mounted Intel RealSense D405 camera.
- Built the data pipeline behind them: aligns 100 Hz robot state logs to 30 Hz camera frames and packages each episode as a LeRobot v3.0 dataset.
- Trained ACT, diffusion policy and $\pi_0 / \pi_{0.5}$ on this data and ran them closed-loop on the arm at 30 Hz, fine-tuning the VLAs with LoRA and NF4/int8 quantization on rented GPUs.
- π_0 reached 80% on sanding (265 episodes, 4.4 h, 1 instruction) and 65% on colour sorting (400 episodes, 5.1 h, 9 instructions), each over 20 autonomous trials. $\pi_{0.5}$ is training now for pick-to-light — retrieve an object from a named rack slot and place it — on a 356-episode, 0.8 h dataset spanning 6 instructions.
- Reimplementing some of the robotic learning policies ACT from the paper - CVAE plus transformer, built up from autoencoder and VAE components.

AnimalCLEF 2026: Wildlife Re-identification — 9th Place

Apr 2026 - May 2026

AnimalCLEF 2026 Challenge

- Performed unsupervised clustering of ~2,400 unlabeled images across 4 species (lynx, salamanders, sea turtles, Texas horned lizards).
- Fused global embeddings (MiewID v3 + MegaDescriptor) with local feature matching (KAZE, ALIKED+LightGlue).
- Fine-tuned MegaDescriptor with ArcFace loss for a +6.4% gain; used SAM2 to mask background keypoints before matching.
- Ran 30+ documented experiments; secured 9th place on the leaderboard.

GIV-CXR: Densely Grounded, Visually Interpretable, Chest X-Ray Question

Aug 2024 - Jan 2025

Answering Dataset

VLaNC Lab, Mahindra University (Dr. Nidhi Goyal) · in collaboration with IIIT Hyderabad

- Curated a densely grounded chest X-ray VQA benchmark with 355,293 question-answer pairs linked to bounding-box annotations across 36 anatomical regions, enabling systematic spatial-reasoning evaluation.
- Built a hallucination-filtering pipeline combining automated detection with radiologist validation, achieving an 82.4% acceptance rate (18.2% hallucination rejection).
- Developed grounded VLM baselines; best model achieved 68.1 mIoU for spatial grounding (nearly 2x GPT-4o-mini's 39.0) while leading on G-Eval answer quality (3.98).

Enhancing video summarization by enhancing text-to-image module

Aug 2023 - Dec 2023

Supervisors: Dr. Nidhi Goyal, Assistant Professor, Mahindra University

- Improved text-to-image projection by augmenting it with a learnable projection layer.
- Finetuned the selected baseline's projection module using the QVHighlights dataset.

Work Experience

Senior Data Scientist

May 2026 - Present

Penguin AI, Hyderabad, India

- Building end-to-end LLM-based agent pipelines for healthcare workflows, incorporating live production user feedback for self-improving, data-driven agent development.
- Leading AI governance efforts — compliance, bias mitigation, and safety alignment across deployed systems.
- Designing ML system architecture for cost optimization and scalability of agent infrastructure in production.

Data Scientist

Jun 2025 - May 2026

Penguin AI, Hyderabad, India

- Built production-grade LLM workflow systems to automate prior authorization, reducing medical nurse review time by approximately 30-40% with around 80-85% accuracy; currently improving the system with a HITL workflow.
- Working on applied research methodologies to enhance automated medical coding.
- Developed an ML system for appeal denial prediction directly from customer data sources, with a corresponding ETL pipeline.

Associate Data Scientist

Jun 2024 - Jun 2025

Awone AI, Hyderabad, India

- Developed an end-to-end interruptible voice chat application with open-source components with MCP client-server for a user database.
- Improved the inference speed of production LLMs by 15% by quantization and developing speculative decoding n-gram head.
- Built and deployed a BERT-based NER model for healthcare PII redaction using a custom loss function.

Data Science Intern

Jan 2024 - Jun 2024

Awone AI, Hyderabad, India

- Implemented Llama-2 and Mistral from scratch to study their architectures and attention mechanisms, and applied quantization and other optimization techniques to reduce inference latency.
- Finetuned task-specific SLMs for medical-insurance related tasks.

Research intern

Jul 2023 - Sept 2023

Mahindra University, Hyderabad, India

- Surveyed video summarization approaches, challenges, and future directions with a four-person team, laying groundwork for follow-on modeling work.

Achievements

9th place in AnimalCLEF 2026 Wildlife Re-identification Challenge

2026

4 years consecutive merit scholarship (2020-2024) for securing a rank in the top 5% of the branch

2024

3rd position in ICETCI Hackathon (2023), Mahindra University

2023

Top 150 teams (Rank 111) in Amazon ML Challenge 2023, a total of 4000+ teams participated

2023

Courses & Certifications

Modern Robotics, Course 1: Foundations of Robot Motion

Apr 2026

Coursera — Northwestern University · Grade: 98.46%

Modern Robotics, Course 2: Robot Kinematics

In Progress

Coursera — Northwestern University

Skills

Languages & Libraries

Python, SQL, Pandas, NumPy, Scikit-learn, HuggingFace, PyTorch, OpenCV, C++

AI Engineering & Infrastructure

LangChain, LangGraph, vLLM, MLFlow, Ray

Cloud & Deployment

AWS, Docker, Async programming, Linux, GitHub Actions, MongoDB, PostgreSQL, Kubernetes

Robotics & Simulation

ROS 2, LeRobot, Fairino SDK, SO-101, RealSense, MuJoCo, CoppeliaSim

Extra-Curricular

Power-lifting Volunteer for AERO College Gym committee member (MU)

Coordinated events at Aero 2023 Sports Fest

Oversaw equipment and mentored new students

Yoga club member and volunteer (MU)

Volunteered for the University's celebration of International Yoga Day.